

Data Structures and Algorithms

1. Find out list of companies which are conducting placement drive using DSA
2. What is the syllabus of DSA in GATE?
3. Find out what are set of variables in Amazon invoice bill, gas bill, current bill

→

- username - string
- itemname - string
- price - int
- no of items - int
- phone number - unsigned long int
- address - string
- OTP - unsigned int
- Reviews - string
- Brandname - string
- Gmail - string

→ Ticket number - unsigned int

start - string

Destination - string

Ticket cost - int

Train name - unsigned int

Date - Date

seat number - unsigned int

HW:-

- i. (i) Google
- (ii) Microsoft
- (iii) Amazon
- (iv) Apple
- (v) Meta (Facebook)
- (vi) Netflix
- (vii) Adobe
- (viii) Oracle
- (ix) Flipkart
- (x) Infosys
- (xi) TCS
- (xii) Wipro

- 2.
1. Arrays
 2. Linked lists
 3. stacks
 4. Queues
 5. Trees
 6. Heap
 7. Hashing
 8. Graphs
 9. Graph traversals
 10. Searching algorithms
 11. Sorting algos
 12. Recursion
 13. Divide and conquer
 14. Greedy Alogs
 15. Dynamic programming
 16. Backtracking
 17. Time complexity
 18. Space complexity

3. Variables in amazon invoice bill

orderId	string
Billing address	string
shipping address	string
order date	int
invoice number	string
invoice date	int
PAN no.	unsigned long int
GST	float
Delivery charge	float
payment method	string
phone number	string
Email	string
Total amount	double

→ Gas bill

consumer number	string
Consumer name	string
address	string
meter number	string
prev. reading	int
current reading	int
Tax	float
Date	date

→ current bill

consumer ID	string
consumer name	string
meter number	string
prev. reading	int
current reading	int
units consumed	int
rate per unit	float
total bill	float
due date	date
payment status	boolean

1. Read n input array, delete given input, user input index of an array with -1, continuously ask user the indices to delete the values as long as user choice is "yes" stop this process when user choice is 'no', all the array values are detected

2. Consider an array of size 'n', store the values baring on the value of unit position

Eg:- 32 → stored in 2nd location
56 → stored in 6th
78 → stored in 8th

C++

```
#include <iostream>
using namespace std;
struct student
```

```
{
    int roll;
```

```
    char name[20];
```

```
    float marks;
```

```
    char grade;
```

```
};
```

```
int main()
```

```
{
```

```
    cout << "size: " << size;
```

```
    return 0;
```

```
}
```

size — int — 4 bytes

char(name[20]) = 20×1 bytes

char(grade) = 1 byte

float = 4 bytes

Total = 29 bytes

output: — size: 32 bytes

padding — It extra unused memory used

by the compiler so that data is

stored at memory addresses that

that processor can access efficiently. Padding adds 3 bits to grade (char) so that float, int, short, a 4 bit aligned memory. The address.